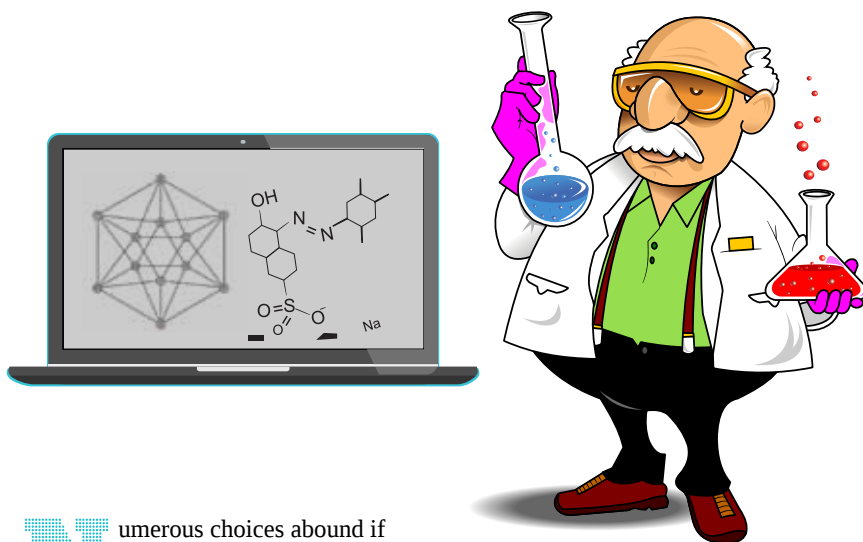


SageMath: The Beginning

SageMath is powerful open source software for scientific computing, and it is important to get familiar with it in these days of AI-based applications. The first part of this series on SageMath will introduce you to its installation and basic usage.



and Singular, as well as Python scientific computing libraries like NumPy, SciPy, among others, readily available? SageMath consolidates numerous specialised CAS software packages into a unified interface that closely resembles Python. As a result, prospective users only need to be familiar with Python syntax, rather than mastering the unique conventions of multiple CAS software systems. Apart from offering this unified interface, SageMath also has several exclusive functions and utilities.

If you are convinced that investing time in learning SageMath is worthwhile, note that while it is easy to start using SageMath, mastering it may take some time. Therefore, I will present a series of articles to cover key topics in the vast realm of SageMath. The series will commence with the installation and basic usage of SageMath, accompanied by simple examples. Subsequently, it will delve into functions associated with various fields of mathematics, including but not limited to graph theory, combinatorics, numerical analysis, number theory, statistics, algebra, linear algebra, calculus, and more. You may be wondering about the relevance of delving into these challenging mathematical topics. However, it's crucial to note that these mathematical concepts serve as the foundation for scientific computing, which, in turn, represents the future of computing. The series aims to discuss the fundamental mathematical theories required, followed by extensive coding examples.

Numerous choices abound if you are interested in learning an advanced computing tool, supplementing your already rich repertoire of traditional and popular programming languages and tools. Nevertheless, I propose an outstanding recommendation for your consideration: SageMath. SageMath is a computer algebra system (CAS) with utilities that span nearly all fields of mathematics. It is free and open source software licensed under GNU GPL. SageMath is very powerful software and an open source alternative to proprietary scientific computing tools like Maple, Mathematica, MATLAB, etc. The initial development of SageMath was undertaken by William Stein, and the inaugural version was released in February 2005. Figure 1 illustrates the SageMath logo.



Figure 1: The logo of SageMath

But let us first address a significant question: Why should students or professionals in the field of computer science consider learning scientific computing software? Is mastering conventional and widely-used programming languages like Python, Java, C++, C, JavaScript, etc, not enough? Well, we must explore software like SageMath due to the dynamic nature of our times. The emergence of AI-driven applications compels us to move beyond traditional programming and conventional software. Today's end users seek software that leverages AI and machine learning techniques, necessitating a strong foundation in scientific computing. In such contexts, software like SageMath becomes indispensable and highly valuable.

Another crucial question is: what sets SageMath apart when there are numerous CAS software and scientific computing tools such as GAP, Maxima,